

Power Cell Escape

Game Concept and Design Documentation

Yuhang Chen

1. Theme and Intended Player Experience

The theme of this game is about a disoriented protagonist regains consciousness in an abandoned, power-failing facility. There is no combat capability, no map, no clear memory of what happened — only the immediate goal of restoring enough power (via 3 power cells) to unlock the door and escape, while monsters are patrolling around the map and try to catch him.

The intended player experience could be:

- Tension — The threat is always a possibility, even off-screen. Low lighting, ambient hums, and audio cues (footsteps, mechanical groans) keep the player alert at all times.
- Exploration — The player searches a single but multi-room layout under restricted visibility to find the three power cells, rewarding attentiveness over speed.
- Vulnerability — The player cannot fight back. Survival depends on route planning, timing, and use of cover — the power imbalance between player and threat is constant and intentional.
- Escape Satisfaction — Once all three cells are delivered and the exit unlocks, reaching it delivers a short, conclusive payoff: a genuine sense of having gotten out against the odds.

2. Core Gameplay Loop

The overall game loop is showing below:

- Explore
- Locate Power Cell
- Evade Threat
- Collect
- (repeat × 3)
- Return to Exit Panel
- Unlock Door
- Escape

The loop is driven by collection-gated escalation: each power cell collected raises the facility's danger level -- the threat's patrol speed, detection range, or the ambient lighting/sound intensity increases slightly. The player becomes simultaneously closer to freedom and more at risk.

In addition, the threat, which are the monsters in this game, function as a pressure system, not a combat encounter. Players have no any capability to eliminate or fight against the monsters but could only hide and escape. These monsters move at a competitive but escapable speed and has limited turning/vision, forcing the player to use geometry — corners, obstacles, hiding spots — to avoid it rather than confront it.

3. Rules and Controls

Rules:

- The exit door is permanently locked until all 3 power cells are collected (0/3 → 3/3 tracked by UI counter).
- The player has no attack/combat capability towards any kind of monsters and the only responses to the threat are avoidance, distance, and use of cover.
- Being detected/caught by the monsters, or the fail condition timing out, ends the run immediately.
- Danger level escalates in fixed steps after each cell pickup (1st cell → step 1, 2nd → step 2, 3rd → step 3) The escalation is one-directional and does not reset until the run restarts.
- The run can be restarted at any time from the UI and there is no save and load functionality.
- No resource management is required from the player: There is no stamina, flashlight battery, or noise meter in the current scope; sprinting (Shift) has no cost beyond exposing the player to a slightly wider detection window from the threat (faster footstep sound = easier to be heard/found).
- One monster is active per danger stage to keep the FSM workload predictable within the 7-day scope. Only one single monster patrols per level at any given danger stage (its speed/detection range changes on each stage rather than adding additional monsters). This is a deliberate scope-control decision.

Controls (PC, keyboard):

Input	Action
W / A / S / D	Move
Mouse	Look / Camera Rotation
E	Interact (pick up cell / use exit panel)
Shift	Sprint / Run
ESC	Pause menu (volume control lives here — accessibility)

4. Win and Lose Conditions**Win Condition:**

Collect all 3 power cells → return to the exit control panel → interact to unlock the door → reach/exit through the door → Win Screen displayed, with a restart option.

Lose Condition (either):

- Caught: the monsters reaches the player directly → Players are killed by monsters → Lose Screen displayed, with a restart option.
- Time depleted: the timer above the screen reaches zero → The door closed permanently → Lose Screen displayed, with a restart option.

Both screens are clear, immediate, and always offer a one-button restart.

5. Level Sketch / Diagram



The initial map / level diagram is above. The point is that I may modify this map, which is subjected to the actual development process.

Given that there is only 7 days for this game. I try to make everything as simple as possible. To finish this game in time is the goal.

6. Principal Gameplay and Programming Systems

The initial designed gameplay system may be as followed, **which is subjected to the actual development process:**

System	Key Scripts	Description
Player System	PlayerController.cs	Movement, collision, input
Interaction System	InteractionSystem.cs, Interactable.cs	Proximity detection, key-press trigger, prompt display
Collection System	PowerCellManager.cs	Tracks 0–3 collected, updates UI, gates exit
Threat AI	ThreatFSM.cs, PatrolController.cs, DetectionSystem.cs	Patrol → Detect → Chase finite state machine

System	Key Scripts	Description
Exit / Win Logic	ExitPanel.cs, WinLoseManager.cs	Checks cell count, unlocks door, triggers win/lose screens
UI System	ObjectiveUI.cs, CellCounterUI.cs, InteractionPromptUI.cs	Objective display, counter, prompts
Audio System	AudioManager.cs	Ambient loops, event SFX, threat audio, volume control
Game State	GameStateManager.cs	Global flags, danger-escalation stage, restart handling

If anything change with the systems above, I would highlight it in the github game development log.

7. A Realistic Plan for the Seven-Day Development Period

The initial development plan for the seven days is listed below:

Day	Core Objective
Day 1	Set up the base project framework; implement basic player movement and scene loading
Day 2	Implement a placeholder exit/door system; reserve the interface for power cell key verification logic
Day 3	Implement and extend the AI enemy's movement, tracking, and pursuit logic; add a death state and a restart function
Day 4	Implement the power cell (key) logic, player HUD, and a basic opening/menu UI
Day 5	Add sound effects and particle effects
Day 6	Polish visual/audio effects and add more refined character and environment models + Schedule Play Testing
Day 7	Testing and bug fixing

Threat AI state transitions (ThreatFSM.cs):

- Patrol → Detect: triggered when the player enters the monster's vision cone (angle threshold, e.g. $\pm 60^\circ$) and is within detection range, and no obstacle blocks line-of-sight (raycast check via DetectionSystem.cs).

- Detect → Chase: triggered once detection confidence/timer accumulates past a threshold (to avoid instant "insta-spot" feeling and give the player a reaction window).
- Chase → Patrol: triggered when the player breaks line-of-sight for a set duration (e.g. 3 – 5 seconds), after which the monster returns to its last known position and resumes patrol.
- Chase → Catch: triggered when distance to player falls below a small contact threshold, which calls WinLoseManager.cs to end the run.

Collection → Exit/Win linkage:

- PowerCellManager.cs fires an event (e.g. OnCellCountChanged) each time a cell is collected, rather than being polled every frame. ExitPanel.cs subscribes to this event to update its "locked/unlocked" state and refresh CellCounterUI.cs, so the exit logic and UI stay in sync without unnecessary per-frame checks.
- GameStateManager.cs listens to the same event to trigger the danger-escalation step (adjusting ThreatFSM.cs speed/detection parameters and AudioManager.cs ambient intensity).

If you want to have access to a more detailed one, please check the Kanban on the github. In addition, any changes would also be highlighted in the development log.

8. The assets and resources I intend to use

I try to use all the assets that I use before because there are only 7 days to finish everything. Therefore, finding assets shouldn't take too much time.

All Unity Asset Store assets are used under the Unity Asset Store End User License Agreement (EULA), which permits use in game projects but prohibits redistribution of raw asset files.

- Main Environment — Modular Abandoned Hospital

Asset Name: Modular Abandoned Hospital / Horror Hospital

Creator: Hivemind (Name of creator in Unity Asset Store)

Source:

<https://assetstore.unity.com/packages/3d/environments/fantasy/modular-abandoned-hospital-h>

[ror-hospital-abandoned-hospital-ho-276792](https://assetstore.unity.com/packages/3d/environments/fantasy/modular-abandoned-hospital-h)

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Used as-is for main scene environment

- Humanoid CharacterAsset Name: Fantasy Humanoid Character

Creator: CH Assets (Name of creator in Unity Asset Store)

Source:

<https://assetstore.unity.com/packages/3d/characters/humanoids/humans/hospital-patient-character-20058>

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Used as player character model

- Locker / Cabinet Prop

Asset Name: Locker

Creator: VIS Games (Name of creator in Unity Asset Store)

Source: <https://assetstore.unity.com/packages/3d/props/locker-56405>

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Placed as interactive prop in scene

- Electric Fuse Box Prop

Asset Name: Sockets and Fuse Box Pack

Creator: TrueBlue Studios (Name of creator in Unity Asset Store)

Source: <https://assetstore.unity.com/packages/3d/props/sockets-and-fuse-box-pack-148620>

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Used as environmental prop

- Furniture Items

Asset Name: Korea Craft Design Furniture Item

Creator: Korea Craft & Design Foundation (Name of creator in Unity Asset Store)

Source:

<https://assetstore.unity.com/packages/3d/props/furniture/korea-craft-design-furniture-item-252177>

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Used as room decoration props

- Monster / Creature Model

Asset Name: Creature Horror Mutant (Name of creator in Unity Asset Store)

Creator: Vadim Ziambetov (Name of creator in Unity Asset Store)

Source:

<https://assetstore.unity.com/packages/3d/characters/creatures/creature-horror-mutant-113565>

License: Unity Asset Store EULA — <https://unity.com/legal/as-terms>

Accessed: 2026-08-01

Modifications: Used as main enemy character

- Asset Name: Horror Creature 01

Creator: brvhrtz

Source: <https://pixabay.com/sound-effects/horror-creature-01-brvhrtz-232041/>

License:

Pixabay Content License (Free for use) — <https://pixabay.com/service/license-summary/>

Accessed: 2026-08-01

Modifications: Used as main enemy character

- Monster Sound Effects

Modifications: Used as monster ambient sound

- Background Music — Horror Chase Chapter 1

Asset Name: Horror Chase Chapter 1

Creator: Steve Oxen / Fesliyan Studios Inc.

Source:

<https://www.fesliyanstudios.com/royalty-free-music/downloads-c/chase-scene-music/36>

License: Fesliyan Studios Royalty-Free License (free for non-commercial use; credit required if no donation made) — <https://www.fesliyanstudios.com/policy>

Accessed: 2026-08-01

Modifications: Used as chase scene background music

9. Legal, Ethical, Social, Accessibility, and Security Considerations

- Legal — In this game development process, I only use free/open-source/original assets used, with full licence documentation in README and final report. There is no copyrighted third-party content.
- Ethical — Some tension content avoids gore, graphic violence, or exploitation of real-world trauma. In addition, the players are warned if they are under 18 years old. The fear of the game is generated through atmosphere/audio/AI behaviour instead of the cheap jump scares.
- Social — Age guidance (18+) would be displayed on title screen. This game aims for a controlled, short tense experience rather than sustained distress.
- Accessibility — **I try to implement at least one of the feature below:**
 1. Adjustable volume — primarily supports players who are sensitive to sudden loud audio cues (e.g. jump-scare-adjacent SFX) or who play in shared/quiet environments, letting them tune tension level without losing the game's core audio feedback.
 2. Interaction prompts and objective text — ensures players who are Deaf or hard of hearing, or who play with audio muted, still receive all functionally necessary information (what to interact with, current objective), rather than relying on ambient audio cues alone.
 3. High-contrast highlighting on interactable objects, not colour-only cues — specifically supports players with colour vision deficiency (colour blindness), who may not reliably distinguish colour-only indicators; contrast/shape-based highlighting keeps interactable objects identifiable regardless of colour perception.
- Security — No user data collected in this game. This game is fully local, no network connection and external / untrusted scripts executed.

10. Main Risks to Successful Completion / How to manage them

Here are some concerns about this seven-day game development process

Risk	Likelihood	Impact	Mitigation
Monster AI takes longer than planned	High	High	Build the simplest working FSM first (Day 3) and polish it only after it functions. Keep detection logic to basic distance / line-of-sight, instead of very complicated path finding mechanism
Level layout/art takes too long	Medium	Medium	Use primitive / placeholder geometry first and then add visual polish only after gameplay is fully functional
Insufficient time for external play testings	Medium	Medium	Schedule play testers explicitly on Day 6, not left until Day 7
GitHub commit history doesn't reflect real progress across the 7 days	Medium	High (professionalism grade)	Commit at the end of every working session with descriptive messages. Update Development Log file on the github almost everyday
Scope control to avoid the lacking of time	Medium	High	Treat the Must/Should/Could list as a hard boundary. Any extra idea goes into "Could Have" and would only consider after Day 5
Audio/lighting escalation feels unbalanced (too easy/too hard)	Low	Medium	Reserve Day 5–6 explicitly for balance pass based on internal + external play test feedback

If the FSM (Patrol → Detect → Chase → Catch) is still unstable by the end of Day 3, I would downgrade it to a fixed patrol-path monster with a simple proximity-trigger "catch" radius (no dynamic chase behaviour).

This sacrifices some pursuit realism but guarantees the core loop (exploration + evasion + collection + escape) remains fully playable and testable, protecting the "final game quality" grade weighting even under time pressure.